Page Layout and Document Styling

LaTeX Learning Module

November 12, 2026

1 Choosing a Page Style

The `\pagestyle` command controls the header and footer style used on pages throughout a document. LaTeX includes several built-in choices:

- `plain` places the page number at the bottom of the page.
- `empty` removes the standard header and footer.
- `headings` displays running headings and page numbers.

Use `\thispagestyle{empty}` when a change should affect only the current page. Use `\pagestyle{empty}` when it should apply to the whole document.

2 Setting Margins with geometry

The `geometry` package makes it straightforward to set page dimensions. For example, the following preamble option creates equal margins:

```
\usepackage[a4paper,margin=25mm]{geometry}
```

You can also set each edge separately:

```
\usepackage[
  top=25mm,
  bottom=25mm,
  left=30mm,
  right=20mm
]{geometry}
```

Choose margins that leave enough room for the text and any printed or binding requirements.

3 Creating Headers and Footers

The `fancyhdr` package allows different information to appear in the left, center, and right areas of the header and footer.

```
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{}
\lhead{Course Notes}
\rhead{Page Layout}
\cfoot{\thepage}
```

The commands `\lhead`, `\chead`, and `\rhead` set the header areas. Their footer equivalents are `\lfoot`, `\cfoot`, and `\rfoot`. The command `\fancyhf{}` clears existing header and footer fields before new ones are assigned.

4 Paragraph Indentation and Spacing

LaTeX indents the first line of many paragraphs by default. Set the indentation length in the preamble with `\parindent`:

```
\setlength{\parindent}{0pt}
```

The distance between paragraphs can be adjusted with `\parskip`:

```
\setlength{\parskip}{6pt}
```

Use these settings consistently. Large paragraph gaps combined with first-line indentation may make a page look uneven.

5 Line Spacing

The `setspace` package provides convenient commands for changing line spacing:

```
\usepackage{setspace}
\singlespacing
\onehalfspacing
\doublespacing
```

You can apply a spacing choice to the whole document or use the `spacing` environment for a selected portion:

```
\begin{spacing}{1.25}
This paragraph uses one-and-a-quarter spacing.
\end{spacing}
```

This paragraph is included as a visual example of the wider line spacing enabled by `\onehalfspacing`. It can be useful for drafts that need room for comments or review.

6 Horizontal and Vertical Space

Use `\hspace` to add horizontal space and `\vspace` to add vertical space.

```
Text\hspace{8mm}More text

\vspace{5mm}
A new line after some vertical space.
```

Manual spacing is useful for small adjustments, but avoid using it as the main method for arranging an entire page. LaTeX's layout tools usually produce more reliable results.

7 Page Breaks

The commands `\newpage` and `\pagebreak` both request a new page. They differ in how strongly they encourage the break:

- `\newpage` ends the current page immediately.
- `\pagebreak` suggests a page break and can take an optional strength value from 0 to 4.

For example, `\pagebreak[3]` requests a strong break without making it quite as absolute as `\newpage`.

8 Customizing Headings, Colors, and Lists

The `sectsty` package can change the appearance of section headings. For example:

```
\usepackage{sectsty}
\sectionfont{\color{blue}}
\subsectionfont{\color{teal}}
```

The `xcolor` package provides named colors and lets you define your own. The `\pagecolor` command changes the background color of a page:

```
\usepackage{xcolor}
\pagecolor{yellow!8}
```

Use a light background when the document is intended for reading or printing. To return to the normal background, use `\pagecolor{white}`.

The `enumitem` package gives more control over lists. For example:

```
\usepackage{enumitem}
\setlist[itemize]{itemsep=4pt,leftmargin=18mm}
```

You can also change the settings for one list without changing every list in the document:

```
\begin{itemize}[itemsep=6pt]
  \item First point
  \item Second point
\end{itemize}
```

9 Keeping Text Together with mbox

The `\mbox` command keeps its contents on one line, preventing LaTeX from breaking the text at spaces inside the box.

```
The product code is \mbox{AB 2048 Z}.
```

If the complete box cannot fit on the line, it may extend into the margin. Use this command only for short content that should remain together.

10 Avoiding Page Splits with samepage

The `samepage` environment asks LaTeX to keep a group of material on one page rather than splitting it across a page boundary.

```
\begin{samepage}
A short heading

A paragraph that should stay with the heading.
\end{samepage}
```

This is useful for compact groups of related content. It does not guarantee that an oversized block will fit on a page.

11 Creating Text Blocks with parbox

The `\parbox` command places paragraph text inside a box with a chosen width. LaTeX wraps the text to fit that width.

```
\parbox{0.7\textwidth}{
  This text is placed inside a box that uses
  seventy percent of the current text width.
}
```

A `\parbox` can be useful for short notes, side-by-side text areas, or small formatted blocks.

12 Adjusting Vertical Position with raisebox

The `\raisebox` command moves content up or down relative to the surrounding text baseline.

```
Normal text \raisebox{2pt}{raised text}
and \raisebox{-2pt}{lowered text}.
```

Positive values raise the content; negative values lower it. Small adjustments are usually easiest to read.

13 Manual Line Breaks

Use \\ to end a line where a line break is appropriate, such as in an address or a short title block. The `\newline` command can also create a line break.

```
North Street 12\\
Central District\\
Example City
```

For ordinary paragraphs, leave a blank line in the source to start a new paragraph instead of inserting repeated manual line breaks.

14 Key Points

Page styles control running information, while page-layout settings determine the space available for content. Packages such as `fancyhdr`, `sectsty`, and `enumitem` provide additional control over the document's appearance. Text-positioning commands are best used selectively, so the layout remains clear and consistent.